School of Computer Science Engineering and Information Systems

Winter Semester 2025-26

M. Tech Integrated Software Engineering

ISWE406P-Agile Development Process and DevOps

Fat exam

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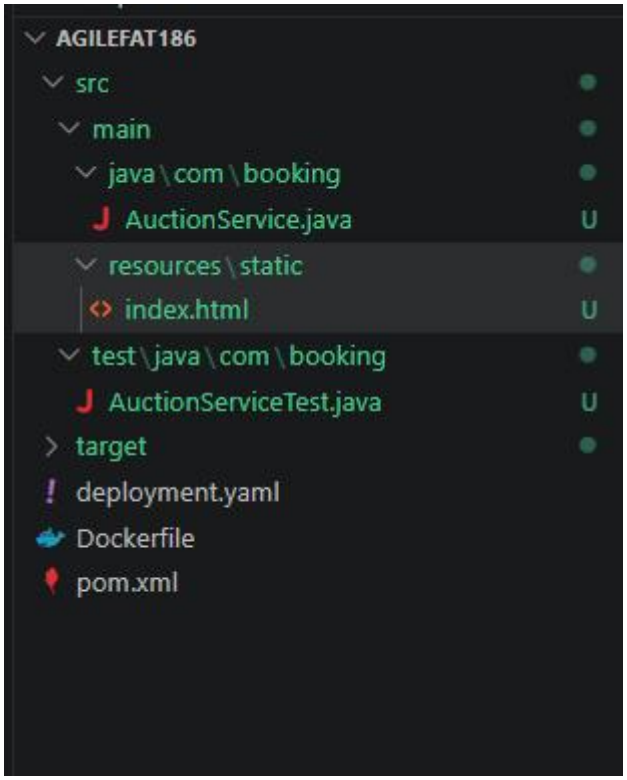
Online auction system

Step1

Installed Docker Desktop, enabled Kubernetes, enabled three plugins (Docker, Docker Pipeline, Kubernetes CLI) .

Step 2 - Create Simple Java Application (in local machine)

The folder structure is



AuctionService.java

```
package com.booking;

import java.util.*;
```

```
public class AuctionService {
```

```java
    // Store bids per item
    private final Map<Long, List<Bid>> bids = new HashMap<>();
```

```java
    // Store starting price
    private final Map<Long, Double> startingPrice = new HashMap<>();
```

```java
    // Add item
    public void addItem(Long itemId, double price) {
        startingPrice.put(itemId, price);
        bids.putIfAbsent(itemId, new ArrayList<>());
    }
```

```java
    // Place bid
    public String placeBid(Long itemId, String bidder, double amount) {
```

```java
        if (!startingPrice.containsKey(itemId)) {
            throw new IllegalArgumentException("Invalid item ID.");
        }
```

```java
        if (bidder == null || bidder.trim().isEmpty()) {
            throw new IllegalArgumentException("Bidder name cannot be empty.");
        }
```

```java
        if (amount <= 0) {
            throw new IllegalArgumentException("Bid amount must be greater
than zero.");
        }
```

```java
        List<Bid> itemBids = bids.get(itemId);
```

```java
        double currentMax = startingPrice.get(itemId);
```

```java
        if (!itemBids.isEmpty()) {
            currentMax = itemBids.stream()
                    .mapToDouble(b -> b.amount)
                    .max()
                    .orElse(currentMax);
        }
```

```java
        if (amount <= currentMax) {
            throw new IllegalArgumentException(
                    "Bid must be greater than current highest bid or starting
price.");
        }
```

```java
        Bid bid = new Bid(itemId, bidder, amount);
        itemBids.add(bid);
```

```java
        return "Bid placed successfully by " + bidder + " for amount " +
amount;
    }
```

```java
    // Get highest bid
    public Bid getHighestBid(Long itemId) {
        List<Bid> itemBids = bids.get(itemId);

        if (itemBids == null || itemBids.isEmpty()) {
            return null;
        }

        return itemBids.stream()
                .max(Comparator.comparingDouble(b -> b.amount))
                .orElse(null);
    }

    // Inner Bid class (so no extra file)
    public static class Bid {
        Long itemId;
        String bidder;
        double amount;

        public Bid(Long itemId, String bidder, double amount) {
            this.itemId = itemId;
            this.bidder = bidder;
            this.amount = amount;
        }

        public String getBidder() {
            return bidder;
        }

        public double getAmount() {
            return amount;
        }
    }
}
```

AuctionServcieTest.java

```java
package com.booking;

import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;

public class AuctionServiceTest {

    private AuctionService service;
```

```java
    @BeforeEach
    void setup() {
        service = new AuctionService();
        service.addItem(1L, 500.0);
    }
```

```java
    @Test
    void validBidShouldPass() {
        String result = service.placeBid(1L, "Alice", 600.0);
        assertEquals("Bid placed successfully by Alice for amount 600.0",
result);
    }
```

```java
    @Test
    void rejectZeroAmount() {
        Exception ex = assertThrows(IllegalArgumentException.class,
                () -> service.placeBid(1L, "Bob", 0));
```

```java
        assertEquals("Bid amount must be greater than zero.", ex.getMessage());
    }
```

```java
    @Test
    void rejectEmptyName() {
        Exception ex = assertThrows(IllegalArgumentException.class,
                () -> service.placeBid(1L, "", 700));
```

```java
        assertEquals("Bidder name cannot be empty.", ex.getMessage());
    }
```

```java
    @Test
    void rejectInvalidItem() {
        Exception ex = assertThrows(IllegalArgumentException.class,
                () -> service.placeBid(99L, "John", 700));
```

```java
        assertEquals("Invalid item ID.", ex.getMessage());
    }
```

```java
    @Test
    void rejectLowerBid() {
        service.placeBid(1L, "Alice", 700);
```

```java
        Exception ex = assertThrows(IllegalArgumentException.class,
                () -> service.placeBid(1L, "Bob", 650));
```

```java
        assertEquals("Bid must be greater than current highest bid or starting
price.",
                ex.getMessage());
    }
```

```java
    @Test
    void highestBidShouldBeCorrect() {
```

```
        service.placeBid(1L, "A", 700);
        service.placeBid(1L, "B", 800);
        service.placeBid(1L, "C", 900);
```

```
        AuctionService.Bid highest = service.getHighestBid(1L);
```

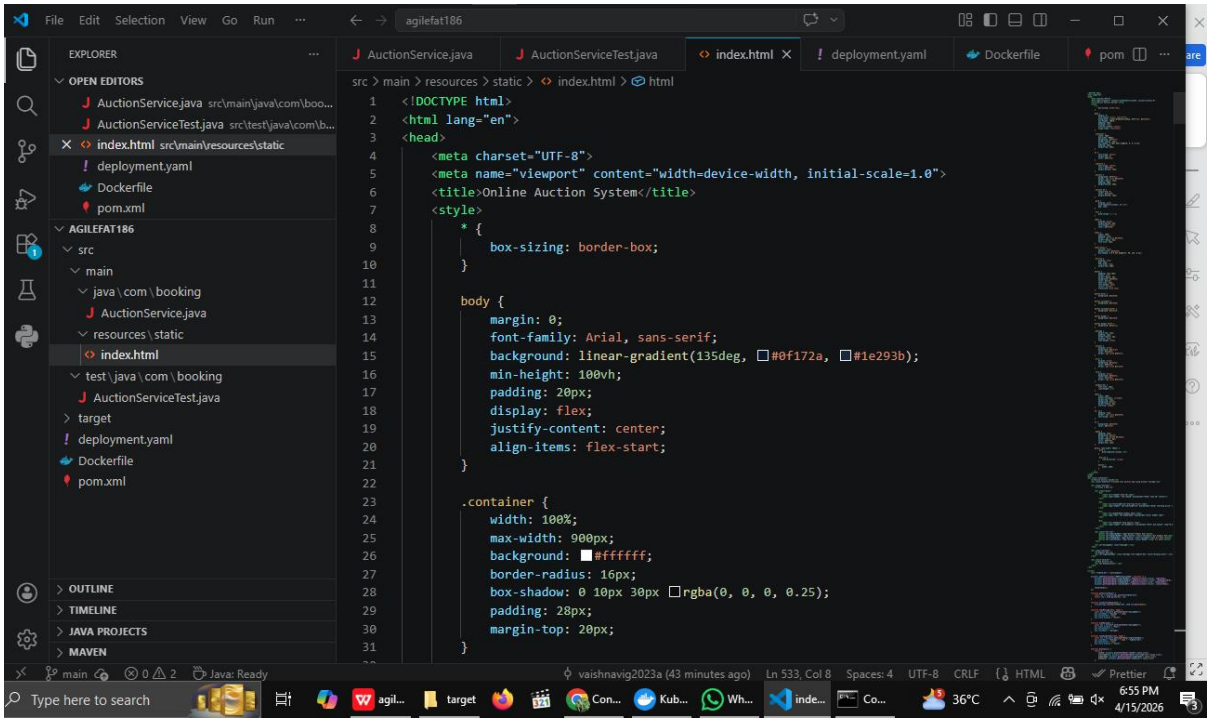
```
        assertNotNull(highest);
        assertEquals("C", highest.getBidder());
        assertEquals(900, highest.getAmount());
    }
}
```

Pom.xml

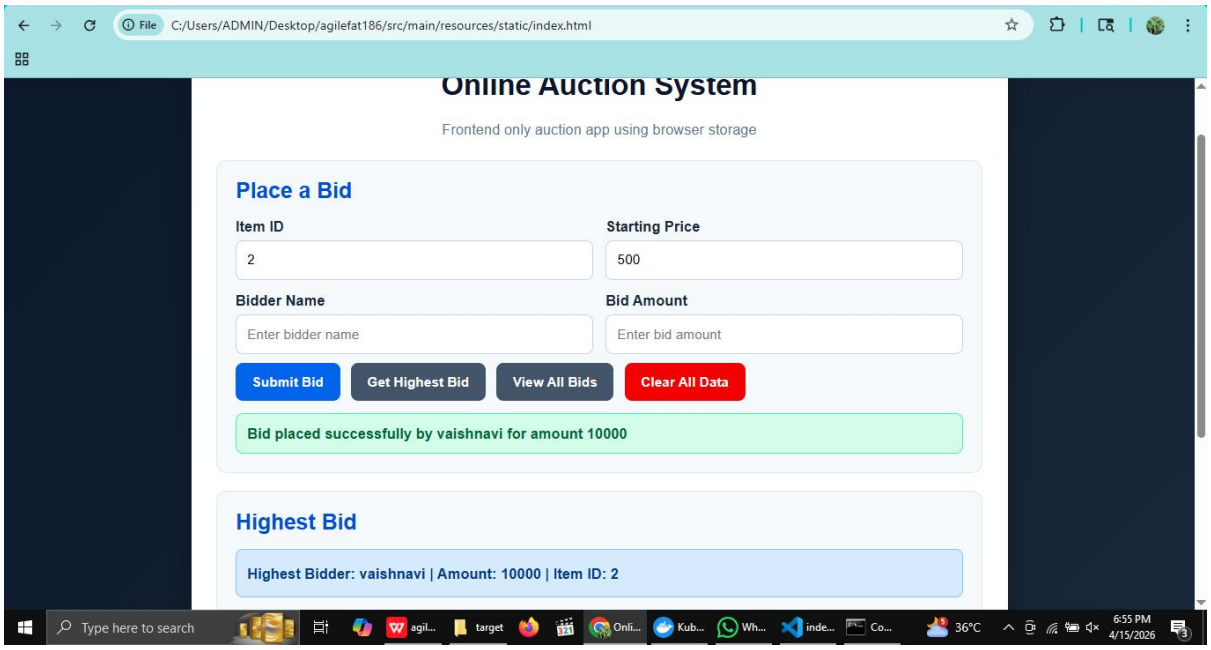
```
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.booking</groupId>
    <artifactId>ticket-booking-app</artifactId>
    <version>1.0</version>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <junit.version>5.10.1</junit.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>${junit.version}</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
            </plugin>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
```

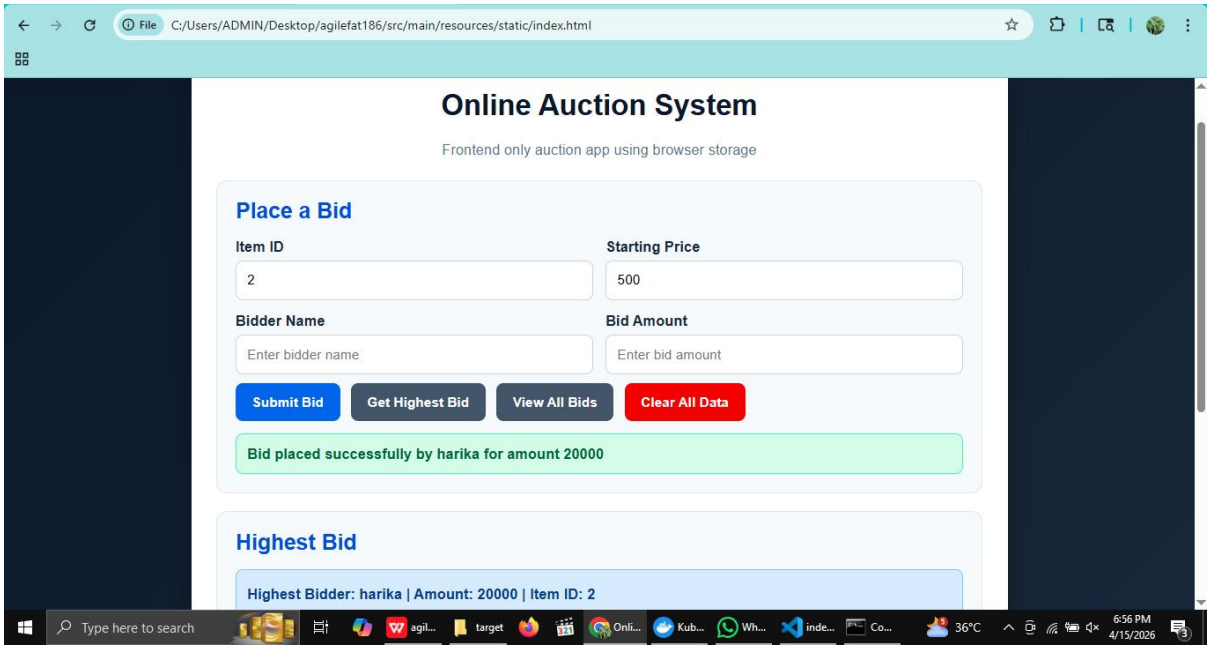
```
            <artifactId>maven-surefire-plugin</artifactId>
            <version>3.2.5</version>
        </plugin>

        <!-- REQUIRED to run jar -->
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-jar-plugin</artifactId>
            <version>3.4.1</version>
            <configuration>
                <archive>
                    <manifest>
                        <mainClass>com.booking.App</mainClass>
                    </manifest>
                </archive>
            </configuration>
        </plugin>

    </plugins>
  </build>
</project>
```

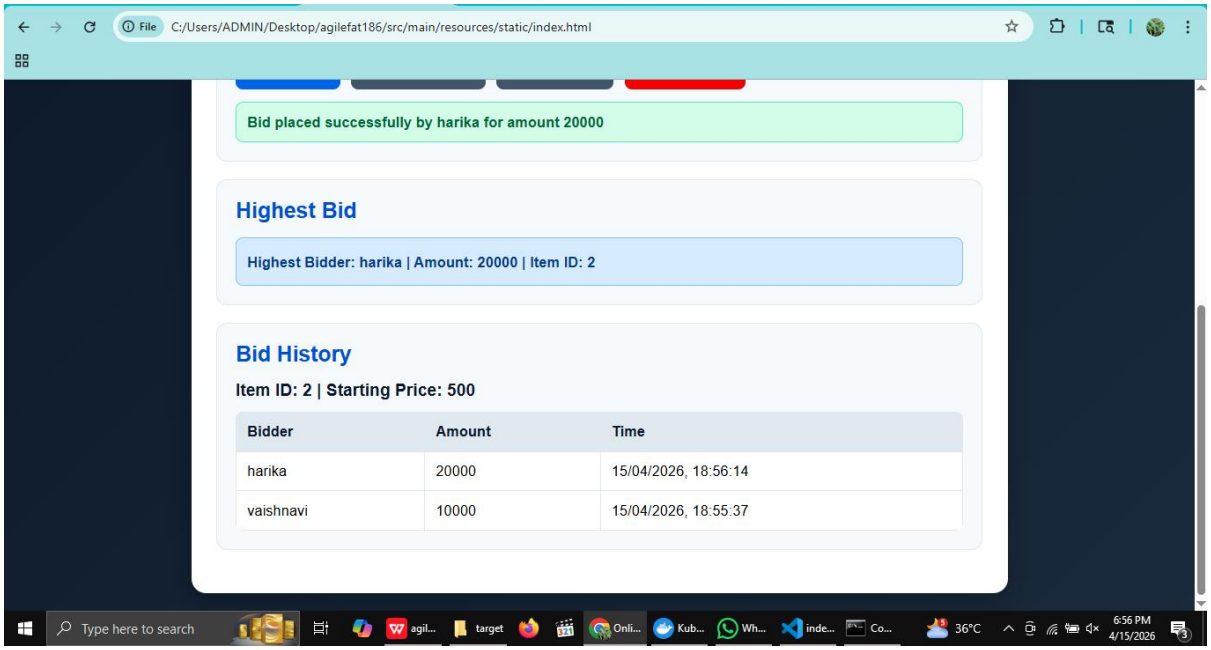
Index.html



Website:


C:/Users/ADMIN/Desktop/agilefat186/src/main/resources/static/index.html
Online Auction System
Frontend only auction app using browser storage
Place a Bid
Item ID
2
Starting Price
500
Bidder Name
Enter bidder name
Bid Amount
Enter bid amount
Submit Bid
Get Highest Bid
View All Bids
Clear All Data
Bid placed successfully by vaishnavi for amount 10000
Highest Bid
Highest Bidder: vaishnavi | Amount: 10000 | Item ID: 2
Type here to search
36°C
6:55 PM
4/15/2026


C:/Users/ADMIN/Desktop/agilefat186/src/main/resources/static/index.html
Online Auction System
Frontend only auction app using browser storage
Place a Bid
Item ID
2
Starting Price
500
Bidder Name
Enter bidder name
Bid Amount
Enter bid amount
Submit Bid
Get Highest Bid
View All Bids
Clear All Data
Bid placed successfully by harika for amount 20000
Highest Bid
Highest Bidder: harika | Amount: 20000 | Item ID: 2
Type here to search
36°C
6:56 PM
4/15/2026

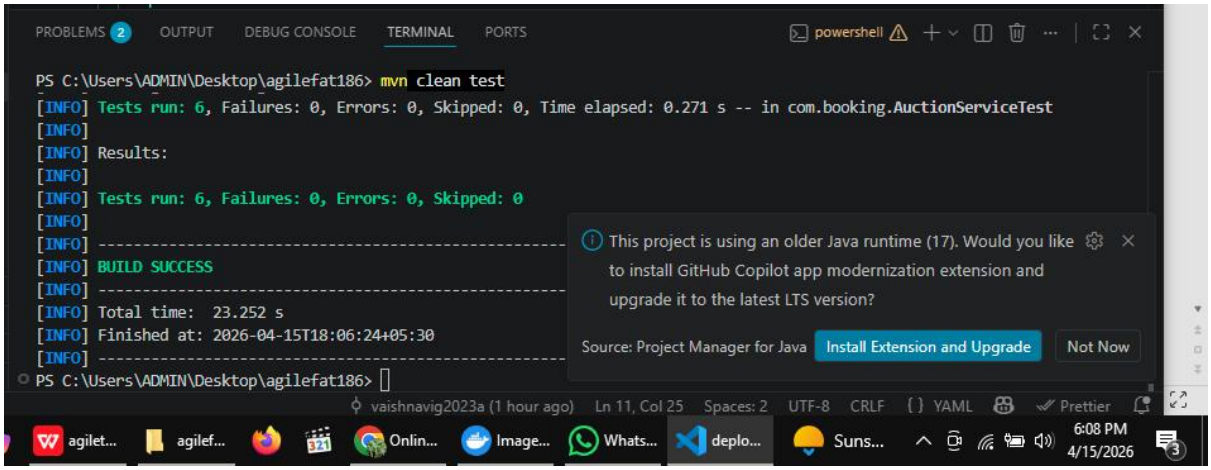


Step 3 - Build Locally using Maven (Test)

Inside project folder:

mvn clean test

mvn clean package



```
PS C:\Users\ADMIN\Desktop\agilefat186> mvn clean package
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ test-java-app ---
[INFO] Building jar: C:\Users\ADMIN\Desktop\agilefat186\target\test-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  15.842 s
[INFO] Finished at: 2026-04-15T18:09:18+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\ADMIN\Desktop\agilefat186>
```

It generates: **target/ test-java-app-1.0.jar**

Name	Date modified	Type	Size
.git	4/15/2026 4:48 PM	File folder	
src	4/15/2026 4:17 PM	File folder	
deployment.yaml	4/15/2026 6:07 PM	Yaml Source File	2 KB
Dockerfile	4/15/2026 4:30 PM	File	1 KB
pom.xml	4/15/2026 4:30 PM	Microsoft Edge H...	3 KB
target	4/15/2026 6:09 PM	File folder	

Name	Date modified	Type	Size
classes	4/15/2026 6:09 PM	File folder	
generated-sources	4/15/2026 6:09 PM	File folder	
generated-test-sources	4/15/2026 6:09 PM	File folder	
maven-archiver	4/15/2026 6:09 PM	File folder	
maven-status	4/15/2026 6:09 PM	File folder	
surefire-reports	4/15/2026 6:09 PM	File folder	
test-classes	4/15/2026 6:09 PM	File folder	
test-java-app-1.0.jar	4/15/2026 6:09 PM	Executable Jar File	8 KB

Step 4 - Create Dockerfile Dockerfile

```
FROM eclipse-temurin:17-jdk-alpine

WORKDIR /app

COPY target/*.jar app.jar

EXPOSE 8080

ENTRYPOINT ["java", "-jar", "app.jar"]
```

Step 5 - Create Kubernetes Deployment File

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: vaishnavi-auction
  labels:
    app: java-test-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: java-test-app
  template:
    metadata:
      labels:
        app: java-test-app
    spec:
      containers:
        - name: java-test-app
          image: sindoori14/agilefat186auction:latest
          imagePullPolicy: Always
          ports:
            - containerPort: 8080
          resources:
            requests:
              memory: "256Mi"
              cpu: "250m"
            limits:
              memory: "512Mi"
              cpu: "500m"
          livenessProbe:
            tcpSocket:
```

```
              port: 8080
            initialDelaySeconds: 30
            periodSeconds: 10
          readinessProbe:
            tcpSocket:
              port: 8080
            initialDelaySeconds: 10
            periodSeconds: 5
---
apiVersion: v1
kind: Service
metadata:
  name: java-test-app-service
spec:
  type: NodePort
  selector:
    app: java-test-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30009
```

Step 6 - Push Code to GitHub

git init

git add .

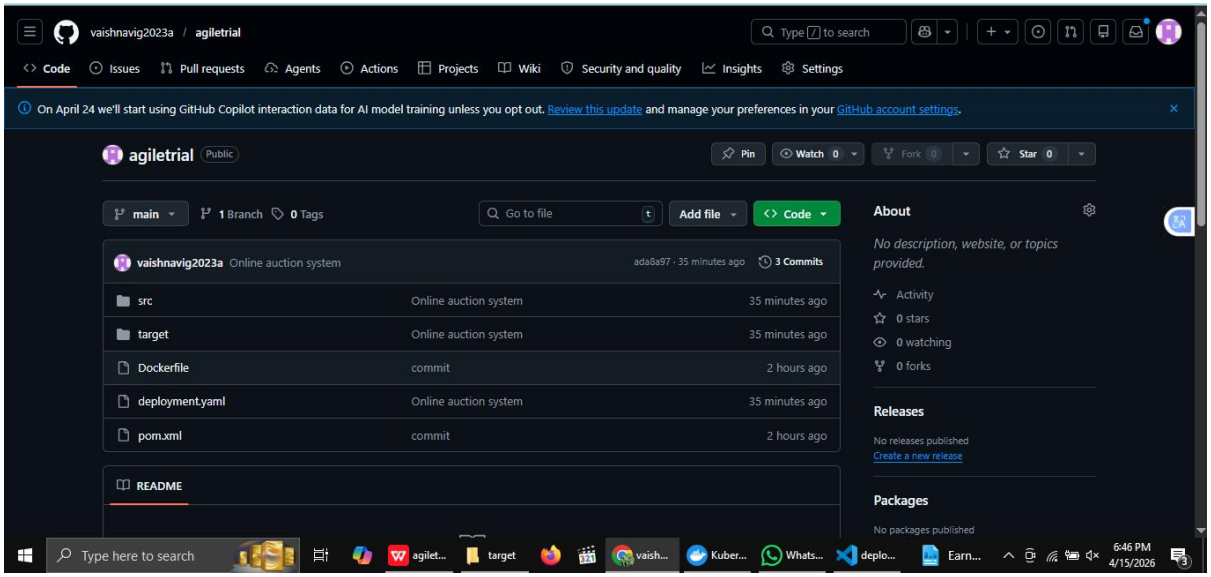
```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                powershell

PS C:\Users\ADMIN\Desktop\agilefat186> git init
Reinitialized existing Git repository in C:/Users/ADMIN/Desktop/agilefat186/.git/
PS C:\Users\ADMIN\Desktop\agilefat186> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will
```

git commit -m "auction system"

```
PS C:\Users\ADMIN\Desktop\agilefat186> git init
Reinitialized existing Git repository in C:/Users/ADMIN/Desktop/agilefat186/.git/
PS C:\Users\ADMIN\Desktop\agilefat186> git add .
PS C:\Users\ADMIN\Desktop\agilefat186> git commit -m "auction system"
On branch main
nothing to commit, working tree clean
PS C:\Users\ADMIN\Desktop\agilefat186> git remote origin main https://github.com/vaishnavig2023a/agiletrial.git
```

git branch -M main
git remote add origin https://github.com/vaishnavig2023a/agiletrial.git
git push -u origin main

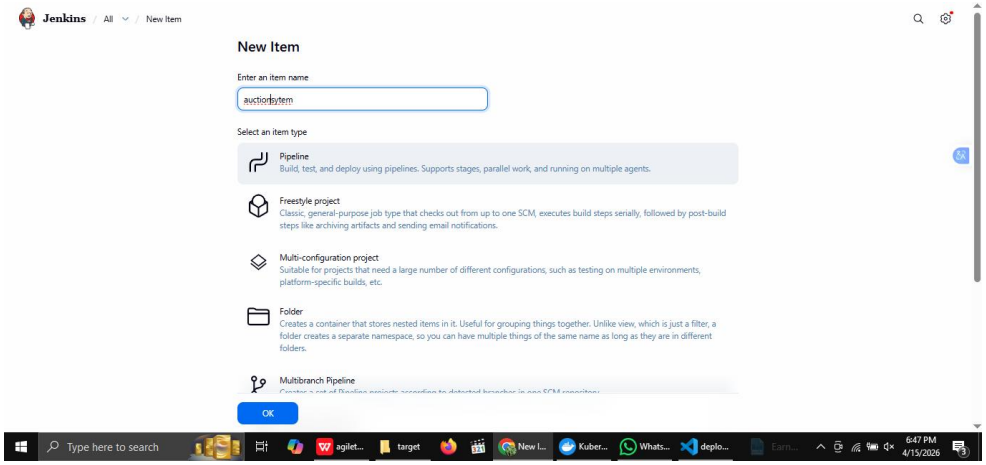
```
PS C:\Users\ADMIN\Desktop\agilefat186> git push origin main
Enumerating objects: 95, done.
Counting objects: 100% (95/95), done.
Delta compression using up to 4 threads
Compressing objects: 100% (55/55), done.
Writing objects: 100% (95/95), 33.21 KiB | 222.00 KiB/s, done.
Total 95 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), done.
To https://github.com/vaishnavig2023a/auction-sytem.git
 * [new branch]      main -> main
```



Step 7 - Create Jenkins Pipeline Job

Run Jenkins

New Item → Pipeline → Name: java-test-app-pipeline

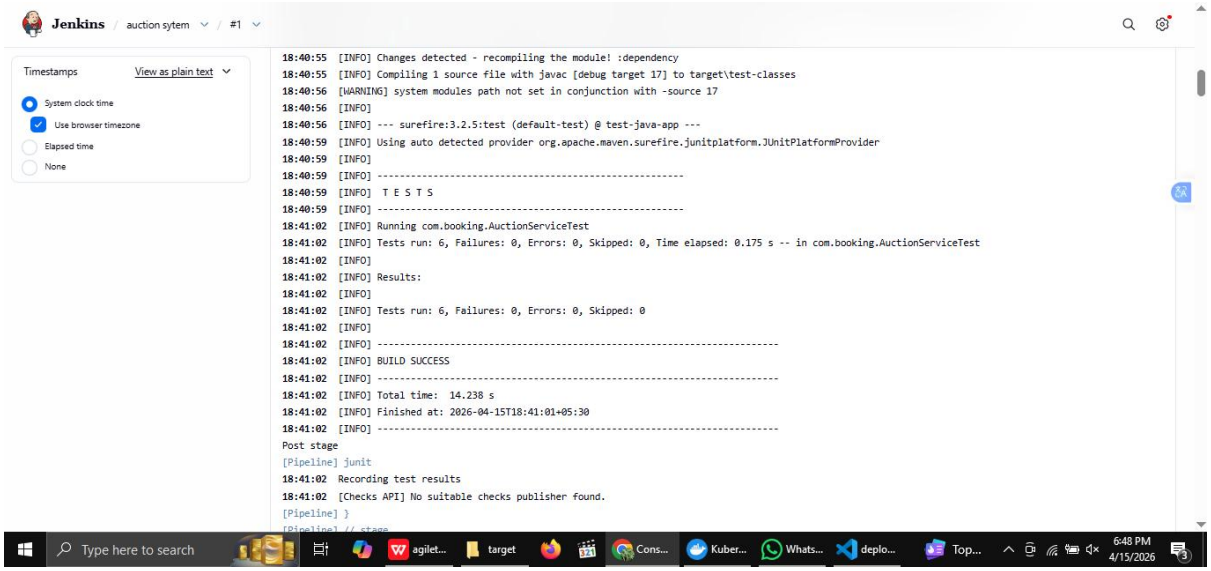


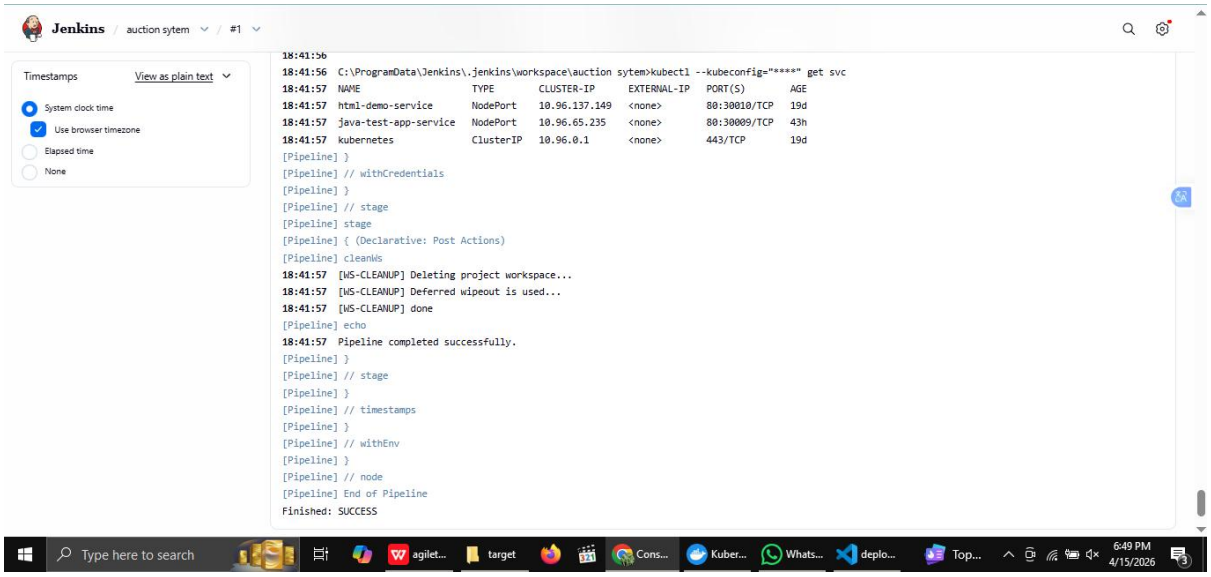
Pipeline Script:

```
pipeline {
    agent any

    options {
        timestamps()
    }

    environment {
        DOCKER_IMAGE = "sindoori14/agilefat186auction:latest"
    }

    stages {
        stage('Clone Repository') {
            steps {
                git branch: 'main',
                    url: 'https://github.com/vaishnavig2023a/agiletrial.git'
            }
        }

        stage('Build & Run Tests') {
            steps {
                bat 'mvn clean test'
            }
            post {
                always {
                    junit allowEmptyResults: true, testResults: 'target/surefire-reports/*.xml'
                }
            }
```

```
}

stage('Package Application') {
    steps {
        bat 'mvn clean package -DskipTests'
    }
}

stage('Build Docker Image') {
    steps {
        bat 'docker version'
        bat "docker build -t %DOCKER_IMAGE% ."
    }
}

stage('Push to DockerHub') {
    steps {
        withCredentials([usernamePassword(
            credentialsId: 'dockerhub',
            usernameVariable: 'DOCKER_USER',
            passwordVariable: 'DOCKER_PASS'
        )]) {
            bat '''
            echo %DOCKER_PASS% | docker login -u %DOCKER_USER% --password-stdin
            docker push %DOCKER_IMAGE%
            '''
        }
    }
}
```

```
        stage('Deploy to Kubernetes') {
            steps {
                withCredentials([file(credentialsId: 'kuberconfig', variable: 'KUBECONFIG')]) {
                    bat '''
                    kubectl --kubeconfig="%KUBECONFIG%" get nodes
                    kubectl --kubeconfig="%KUBECONFIG%" apply -f deployment.yaml
                    kubectl --kubeconfig="%KUBECONFIG%" rollout status deployment/agilemid-
deployment --timeout=180s
                    kubectl --kubeconfig="%KUBECONFIG%" get pods
                    kubectl --kubeconfig="%KUBECONFIG%" get svc
                    '''
                }
            }
        }
    }

    post {
        success {
            echo 'Pipeline completed successfully.'
        }
        failure {
            echo 'Pipeline failed. Check the stage output above.'
        }
        always {
            cleanWs()
        }
    }
}
```

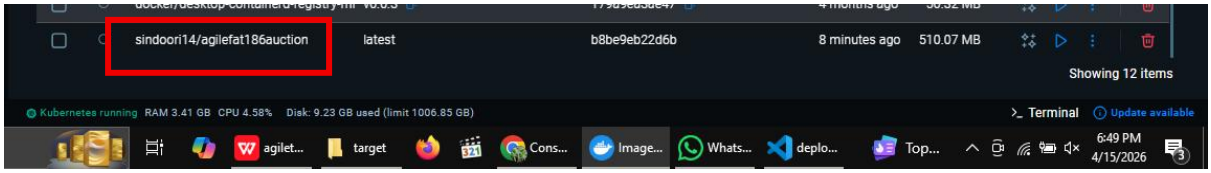


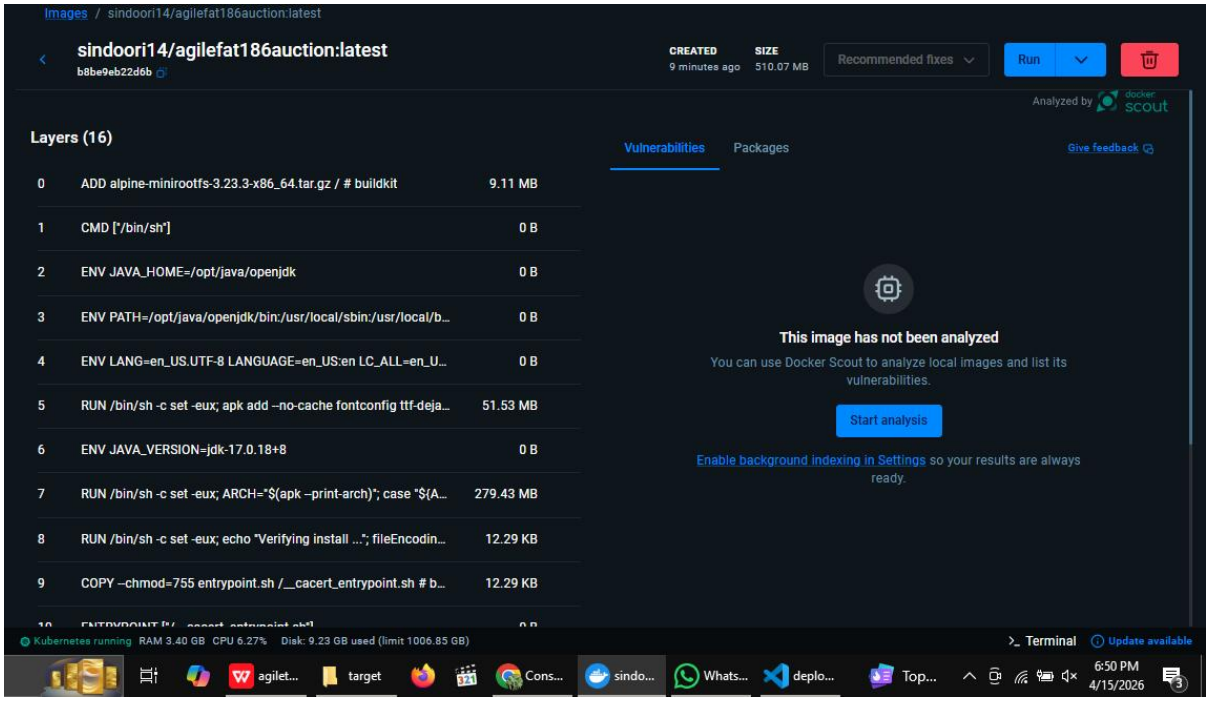


Above console output is the success message

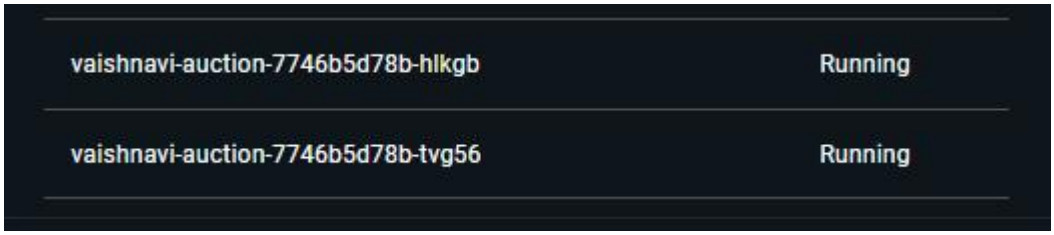
Step 8 - Verify Docker and Kubernetes Deployment

Docker Image





VERIFY KUBERNETES DEPLOYMENT



kubectl get pods

```
C:\Users\ADMIN>kubectl get pods
NAME                                    READY   STATUS             RESTARTS        AGE
agilefat186-vaishnavi-5d58fd4999-7xhrx  0/1     CrashLoopBackOff   23 (3m40s ago)  120m
agilefat186-vaishnavi-5d58fd4999-tj946  0/1     CrashLoopBackOff   23 (3m15s ago)  120m
agilemid-deployment-55b8fdd46c-fg7b2    0/1     CrashLoopBackOff   33 (3m21s ago)  43h
agilemid-deployment-55b8fdd46c-lcffx    0/1     CrashLoopBackOff   33 (3m12s ago)  43h
html-demo-dccf9bf7d-259n9               1/1     Running            4 (68m ago)     13d
html-demo-dccf9bf7d-shd4t               1/1     Running            4 (68m ago)     13d
vaishnavi-auction-7746b5d78b-hlkgb      0/1     CrashLoopBackOff   6 (4m6s ago)    10m
vaishnavi-auction-7746b5d78b-tvg56      0/1     CrashLoopBackOff   6 (4m1s ago)    10m
```

kubectl get deployements

```
C:\Users\ADMIN>kubectl get deployments
NAME                    READY   UP-TO-DATE   AVAILABLE   AGE
agilefat186-vaishnavi   0/2     2            0           121m
agilemid-deployment     0/2     2            0           43h
html-demo               2/2     2            2           13d
vaishnavi-auction       0/2     2            0           10m
```